

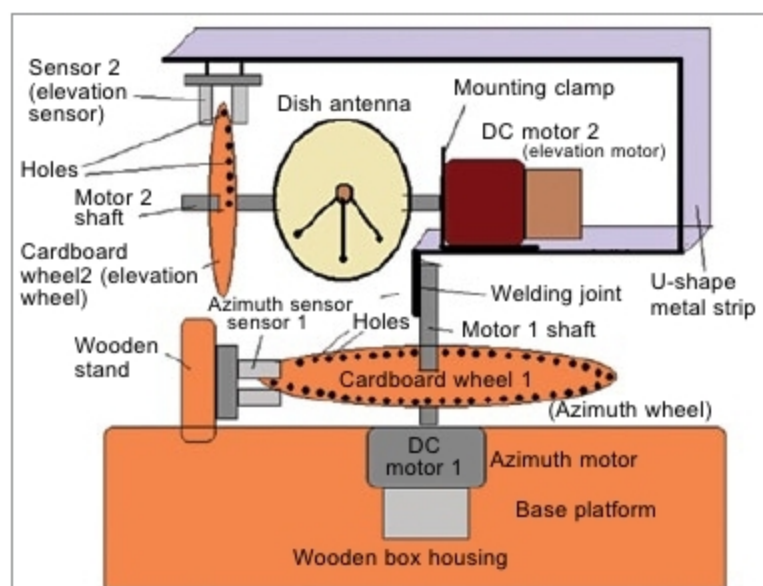


Fig. 3: Proposed satellite dish antenna mounting system

PARTS LIST

Semiconductors:

Board1	- Arduino Uno
IC1, IC2	- MOC7811 sensor
IC3	- L293D motor driver
LCD1	- 20x4 LCD RG2004A or equivalent

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1-R5	- 10-kilo-ohm
R6, R8	- 330-ohm
R7, R9	- 1-kilo-ohm
R10	- 220-ohm

Miscellaneous:

CON1, CON2	- 2-pin terminal connector
S1-S5	- Tactile switch
M1, M2	- 12V DC, 10rpm motor
	- Berg strip

(Except mechanical mounting system, all these are available from kitsnspares.com)

L293D for the motor supply.

Actual controlling of the dish antenna is done using software (dish_contr.ino) written in Arduino programming language and compiled using Arduino IDE software.

Demonstration model

Fig. 3 shows the proposed satellite dish antenna mounting system for a demonstration model. There are two DC geared motors rated at 12V DC with 10rpm speed. The motor that rotates the dish antenna in horizontal plane is azimuth motor 1 (AZ motor or M1), and the one that rotates it in vertical plane is elevation motor 2 (EL motor or M2).

Two opto-interrupter sensors (MOC7811) are used to provide feedback of motor angular positions. Encoding wheels are attached to the shaft of each motor. Each encoding wheel has holes (say, 6mm dia) at 10° , 20° , 30° and so on, up to 350° . The arrangement is done in such a way

that, as the motor rotates, the wheel also rotates and its holes pass through the slotted gap of MOC7811 sensor. Interruption of IR light by each hole generates one pulse per 10° of motor rotation. The proposed arrangement of the mechanical assembly is explained step-by-step below.

1. The entire system is set up in a wooden box that houses the circuit and also serves as the base or platform for demonstration of the model.

2. Azimuth motor (M1) is fixed inside the wooden box using bolt and screws. Encoding cardboard wheel 1 (azimuth) with 35 holes is attached to its shaft.

3. Azimuth sensor 1 (IC1) is also mounted vertically using a small wooden stand on the same wooden platform such that holes of azimuth encoding wheel 1 (azimuth) pass through the slotted gap of azimuth sensor 1.

4. A U-shaped metal strip of around 5cm is directly welded to the shaft of elevation motor (M2). One arm of the U-shaped metal strip is longer than the other arm.

5. Elevation sensor 2 (IC2) is fixed at the end of the longer arm of this metal strip. Elevation motor 2 is mounted on the shorter arm using mounting clamp and screws.

6. A dish antenna made up of cardboard and aluminium foil is directly fixed on the shaft of elevation motor 2. Encoding wheel 2 (elevation wheel) with seven holes is also attached on the shaft of DC motor 2.

7. This encoding elevation wheel 2 is fixed in such a way that holes of the wheel pass through the slotted gap of elevation sensor 2.

Working of demonstration model

Initially, set the angles of both the motors to 0° . This means, the dish antenna has 0° azimuth angle and 0° elevation angle. Azimuth angle can be

set from 0° to 350° in steps of 10° using pushbutton switches S1 and S2.

Then, set elevation angle from 0° to 70° in steps of 10° using S3 and S4.

When both angles are set, press start pushbutton S5 to rotate the dish antenna. Automatically, azimuth motor 1 will start rotating to set the dish antenna in azimuth angle. As AZ motor 1 rotates, encoding wheel 1 will also rotate and its holes will pass through the gap of sensor 1.

The sensor generates pulses that give feedback of the motor's angular position. The motor rotates till the set angle is reached. If set angle is more than the angle of initial position motor 1, it will rotate in clock-wise direction, and if set angle is less than the angle of initial position motor 1, it will rotate in counter-clock-wise direction.

Next, elevation motor 2 will start rotating automatically. It will rotate the dish antenna as well as elevation encoding wheel 2 till set angle is reached.

Once sensor outputs match the entered values of azimuth and elevation angles, both motors will stop rotating and dish antenna will remain in that position. Thus, the dish antenna will attain the desired angular position in azimuth and elevation planes.

If you want to change the dish antenna position (like tracking another satellite), change azimuth and elevation angles by entering new values using input switches to move the dish antenna to a new position.

Operation

Initially, when power is switched on, LCD1 will show AZ angle and set EL angle as 0° . Set AZ angle by pressing AZ angle inc or dec buttons, and EL angle by pressing inc or dec buttons. Every time a button is pressed, the angle changes by 10° .

After setting the angles, press S5. Motor 1 will rotate, followed by motor 2. LCD1 will display the values of azimuth and elevation angles. When the dish antenna attains the desired position, LCD1 will show the set dish position. **EFY**



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EFY Note

The source code of this project is available for free download at source.efymag.com